Linux Commands

1. clear

Description: Clears all previous output from the terminal screen, giving a fresh, empty prompt. It does not delete any files or affect command history.

Syntax: `clear`

Output:

```
isha@Ubuntu:~/Desktop$ clear
isha@Ubuntu:~/Desktop$
```

2. pwd

Description: Prints the absolute path of the current working directory (Print Working Directory).

Syntax: `pwd`

Output:

```
isha@Ubuntu:~/Desktop$ pwd
/home/user/documents
```

3. ls

Description: Lists the files and directories present in the current or specified directory. Common options: -l (long/detailed listing), -a (show hidden files), -h (human-readable sizes).

Syntax: `ls [options] [directory]`

Output:

```
isha@Ubuntu:~/Desktop$ ls
Documents  Downloads  notes.txt  Pictures
```

4. mkdir

Description: Creates one or more new directories. The -p option creates parent directories as needed if they do not already exist.

Syntax: `mkdir [options] directory_name`

Output:

```
isha@Ubuntu:~/Desktop$ mkdir Projects
isha@Ubuntu:~/Desktop$ mkdir -p Projects/App/src
isha@Ubuntu:~/Desktop$
```

5. cd

Description: Changes the current working directory to the specified directory. cd .. moves up one level; cd ~ or cd with no argument moves to the home directory.

Syntax: `cd [directory]`

Output:

```
isha@Ubuntu:~/Desktop$ cd Projects
```

```
isha@Ubuntu:~/Desktop$ pwd
/home/user/Projects
```

6. touch

Description: Creates a new empty file if it does not exist or updates the access and modification timestamp of an existing file without changing its content.

Syntax: `touch filename`

Output:

```
isha@Ubuntu:~/Desktop$ touch report.txt
isha@Ubuntu:~/Desktop$ ls -l report.txt
-rw-r--r-- 1 user user 0 Sep  3 11:20 report.txt
```

7. rm

Description: Removes (deletes) files or directories. -r removes directories recursively, and -f forces deletion without confirmation prompts. Use with caution — deleted files are not sent to a recycle bin.

Syntax: `rm [options] file_or_directory`

Output:

```
isha@Ubuntu:~/Desktop$ rm report.txt
isha@Ubuntu:~/Desktop$ rm -rf OldProject/
isha@Ubuntu:~/Desktop$
```

8. rmdir

Description: Removes empty directories only. If the directory contains any files or subdirectories, the command fails with an error.

Syntax: `rmdir directory_name`

Output:

```
isha@Ubuntu:~/Desktop$ rmdir EmptyFolder
isha@Ubuntu:~/Desktop$
```

9. mv

Description: Moves files or directories from one location to another and can also be used to rename a file or directory by moving it to a new name within the same location.

Syntax: `mv source destination`

Output:

```
isha@Ubuntu:~/Desktop$ mv notes.txt Documents/
isha@Ubuntu:~/Desktop$ mv oldname.txt newname.txt
isha@Ubuntu:~/Desktop$
```

10. cat

Description: Displays (concatenates and prints) the entire contents of one or more files to the terminal. Can also be used to combine multiple files.

Syntax: `cat filename`

Output:

```
isha@Ubuntu:~/Desktop$ cat notes.txt
Meeting at 10 AM
Bring laptop and charger
Submit report by Friday
```

11. cat -n

Description: Displays the contents of a file just like cat, but prefixes every output line with its line number, which is useful for referencing specific lines.

Syntax: `cat -n filename`

Output:

```
isha@Ubuntu:~/Desktop$ cat -n notes.txt
     1     Meeting at 10 AM
     2     Bring laptop and charger
     3     Submit report by Friday
```

12. cp

Description: Copies files or directories from a source location to a destination, leaving the original file intact. The -r option is required to copy directories recursively.

Syntax: `cp [options] source destination`

Output:

```
isha@Ubuntu:~/Desktop$ cp notes.txt notes_backup.txt
isha@Ubuntu:~/Desktop$ cp -r Documents/ Documents_Backup/
isha@Ubuntu:~/Desktop$
```

13. tac

Description: Displays the contents of a file in reverse order — the last line is printed first and the first line is printed last (tac is 'cat' spelled backwards).

Syntax: `tac filename`

Output:

```
isha@Ubuntu:~/Desktop$ tac notes.txt
Submit report by Friday
Bring laptop and charger
Meeting at 10 AM
```

14. head

Description: Displays the first part of a file, by default the first 10 lines. The -n option specifies a custom number of lines to display.

Syntax: `head [-n number] filename`

Output:

```
isha@Ubuntu:~/Desktop$ head -n 2 notes.txt
Meeting at 10 AM
Bring laptop and charger
```

15. tail

Description: Displays the last part of a file, by default the last 10 lines. The -n option sets a custom line count, and -f follows a file live as new lines are appended (useful for log files).

Syntax: `tail [-n number] filename`

Output:

```
isha@Ubuntu:~/Desktop$ tail -n 2 notes.txt
Bring laptop and charger
Submit report by Friday
```

16. history

Description: Displays a numbered list of previously executed commands from the current shell session's command history.

Syntax: `history`

Output:

```
isha@Ubuntu:~/Desktop$ history
  1  pwd
  2  ls -la
  3  mkdir Projects
  4  cd Projects
  5  touch report.txt
```

17. grep

Description: Searches for a specified text pattern inside one or more files and prints the matching lines. Common options: -i (case-insensitive), -r (recursive search), -n (show line numbers).

Syntax: `grep [options] "pattern" filename`

Output:

```
isha@Ubuntu:~/Desktop$ grep "report" notes.txt
Submit report by Friday
```

18. diff

Description: Compares two files line by line and displays the differences between them, showing which lines were added, removed, or changed.

Syntax: `diff file1 file2`

Output:

```
isha@Ubuntu:~/Desktop$ diff notes.txt notes_backup.txt
3c3
< Submit report by Friday
---
> Submit report by Monday
```

19. vi

Description: Opens the Vi text editor to create a new file or edit an existing one directly in the terminal. It operates in modes: Normal mode for navigation/commands, Insert mode (press i) for typing text, and Command-line mode (press :) for saving/quitting (e.g., :wq to save and exit, :q! to quit without saving).

Syntax: `vi filename`

Output:

```
isha@Ubuntu:~/Desktop$ vi notes.txt
```

20. sed

Description: A stream editor used to search, replace, delete, and modify text in files or input streams, typically without opening the file in an editor. The 's' command performs substitution.

Syntax: `sed 's/old_text/new_text/' filename`

Output:

```
isha@Ubuntu:~/Desktop$ sed 's/Friday/Monday/' notes.txt
Meeting at 10 AM
Bring laptop and charger
Submit report by Monday
```

21. apt

Description: The package management tool used on Debian and Ubuntu-based systems to install, update, upgrade, and remove software packages.

Syntax: `sudo apt install package_name`

Output:

```
isha@Ubuntu:~/Desktop$ sudo apt install curl
Reading package lists... Done
Building dependency tree... Done
The following NEW packages will be installed:
  curl
0 upgraded, 1 newly installed, 0 to remove.
Setting up curl ...
curl installed successfully
```

22. yum

Description: The package management tool used on older RHEL/CentOS-based systems to install, update, and remove RPM packages.

Syntax: `sudo yum install package_name`

Output:

```
isha@Ubuntu:~/Desktop$ sudo yum install httpd
Loaded plugins: fastestmirror
Resolving Dependencies
--> Running transaction check
Installing : httpd
Complete!
```

23. dnf

Description: The modern package manager for RHEL/Fedora-based systems, serving as the successor to yum with improved dependency resolution and performance.

Syntax: `sudo dnf install package_name`

Output:

```
isha@Ubuntu:~/Desktop$ sudo dnf install httpd
Last metadata expiration check completed.
Dependencies resolved.
Installing: httpd
Complete!
```

24. sudo

Description: Executes a command with elevated (superuser/root) privileges. It prompts for the current user's password before running the command, provided the user has sudo permissions.

Syntax: `sudo command`

Output:

```
isha@Ubuntu:~/Desktop$ sudo apt update
[sudo] password for user:
Hit:1 http://archive.ubuntu.com/ubuntu jammy InRelease
Reading package lists... Done
```

25. wget

Description: Downloads files from the internet over HTTP, HTTPS or FTP directly from the command line, saving them to the current directory.

Syntax: `wget URL`

Output:

```
isha@Ubuntu:~/Desktop$ wget https://example.com/file.zip
Resolving example.com... 93.184.216.34
Connecting to example.com|93.184.216.34|:443... connected.
HTTP request sent, awaiting response... 200 OK
Saving to: 'file.zip'
file.zip           100%[===================>]  1.2M  --.-KB/s    in 0.4s
'file.zip' saved
```

26. curl

Description: Transfers data to or from a server using various protocols (HTTP, HTTPS, FTP, etc.). Commonly used to make HTTP requests, download files or test APIs. The -O option saves the output with the remote file's name.

Syntax: `curl [options] URL`

Output:

```
isha@Ubuntu:~/Desktop$ curl https://api.example.com/status
{"status": "ok", "uptime": "12h 45m"}
```

27. ping

Description: Tests network connectivity to a host by sending ICMP echo request packets and measuring the response time. Press Ctrl+C to stop.

Syntax: `ping hostname_or_IP`

Output:

```
isha@Ubuntu:~/Desktop$ ping google.com
PING google.com (142.250.premium): 56 data bytes
64 bytes from 142.250.x.x: icmp_seq=0 ttl=115 time=12.4 ms
64 bytes from 142.250.x.x: icmp_seq=1 ttl=115 time=11.8 ms
^C
--- google.com ping statistics ---
2 packets transmitted, 2 received, 0% packet loss
```

28. free

Description: Displays the total, used, and available amount of physical (RAM) and swap memory on the system. The -h option shows values in human-readable units.

Syntax: `free [-h]`

Output:

```
isha@Ubuntu:~/Desktop$ free -h
total    used    free   shared  buff/cache  available
Mem:  7.8Gi   2.1Gi   3.2Gi   220Mi   2.5Gi        5.3Gi
Swap: 2.0Gi      0B   2.0Gi
```

29. top

Description: Displays a real-time, continuously updating view of running processes and system resource usage (CPU, memory). Press 'q' to quit.

Syntax: `top`

Output:

```
isha@Ubuntu:~/Desktop$ top
top - 11:42:10 up 2 days,  3:15,  1 user,  load average: 0.15, 0.10, 0.08
Tasks: 210 total,   1 running, 209 sleeping
%Cpu(s):  3.2 us,  1.1 sy,  0.0 ni, 95.4 id
MiB Mem :   7987.0 total,   3312.4 free
  PID USER   %CPU  %MEM   COMMAND
 1024 user    2.3   1.5   firefox
  512 root    0.5   0.3   systemd
```

30. zip

Description: Compresses one or more files or directories into a single .zip archive file. The -r option is used to include directories recursively.

Syntax: `zip [options] archive_name.zip file_or_folder`

Output:

```
isha@Ubuntu:~/Desktop$ zip -r archive.zip Documents/
```

```
  adding: Documents/ (stored 0%)
  adding: Documents/notes.txt (deflated 32%)
  adding: Documents/report.txt (deflated 28%)
```

31. unzip

Description: Extracts the contents of a .zip archive file into the current directory (or a specified destination).

Syntax: `unzip archive_name.zip`

Output:

```
isha@Ubuntu:~/Desktop$ unzip archive.zip
Archive:  archive.zip
   creating: Documents/
  inflating: Documents/notes.txt
  inflating: Documents/report.txt
```

32. find

Description: Searches for files and directories within a directory hierarchy based on criteria such as name, type, size, or modification time.

Syntax: `find path -name "filename"`

Output:

```
isha@Ubuntu:~/Desktop$ find . -name "*.txt"
./notes.txt
./Documents/report.txt
./Documents/notes_backup.txt
```

33. locate

Description: Quickly searches for files by name using a pre-built index database, making it much faster than find, though results depend on how recently the database was updated.

Syntax: `locate filename`

Output:

```
isha@Ubuntu:~/Desktop$ locate notes.txt
/home/user/notes.txt
/home/user/Documents/notes_backup.txt
```

34. updatedb

Description: Updates the database used by the locate command so that recently created or moved files can be found. Typically requires root privileges and is run periodically or manually.

Syntax: `sudo updatedb`

Output:

```
isha@Ubuntu:~/Desktop$ sudo updatedb
(No output on success; the locate database is refreshed in the background)
```

35. useradd

Description: Creates a new user account on the system. Typically requires root/sudo privileges. The -m option creates a home directory for the new user.

Syntax: `sudo useradd [options] username`

Output:

```
isha@Ubuntu:~/Desktop$ sudo useradd -m john
(No output on success; a new user 'john' and home directory /home/john are created)
```

36. passwd

Description: Sets or changes the password for a user account. Running it without an argument changes the current user's own password.

Syntax: `sudo passwd username`

Output:

```
isha@Ubuntu:~/Desktop$ sudo passwd john
New password:
Retype new password:
passwd: password updated successfully
```

37. groupadd

Description: Creates a new user group on the system, which can then be used to organize permissions for multiple users at once.

Syntax: `sudo groupadd group_name`

Output:

```
isha@Ubuntu:~/Desktop$ sudo groupadd developers
(No output on success; the 'developers' group is created)
```

38. usermod

Description: Modifies an existing user account's properties, such as adding the user to a group (-aG), changing the home directory (-d), or changing the login shell (-s).

Syntax: `sudo usermod [options] username`

Output:

```
isha@Ubuntu:~/Desktop$ sudo usermod -aG developers john
(No output on success; user 'john' is added to the 'developers' group)
```

39. chmod

Description: Changes the read, write, and execute permissions of a file or directory, using either symbolic notation (u/g/o + r/w/x) or numeric (octal) notation.

Syntax: `chmod [permissions] filename`

Output:

```
isha@Ubuntu:~/Desktop$ chmod 755 script.sh
isha@Ubuntu:~/Desktop$ ls -l script.sh
-rwxr-xr-x 1 user user 128 Sep  3 11:20 script.sh
```